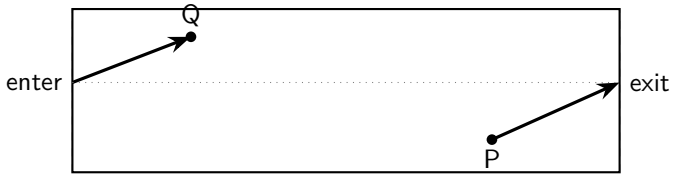
Each yarn route goes upward from A to B, with no self-crossings. Except at A and B, keep the yarn strictly between the rims. To build a different route, lift and re-lay the yarn, putting its ends back on the same marks.

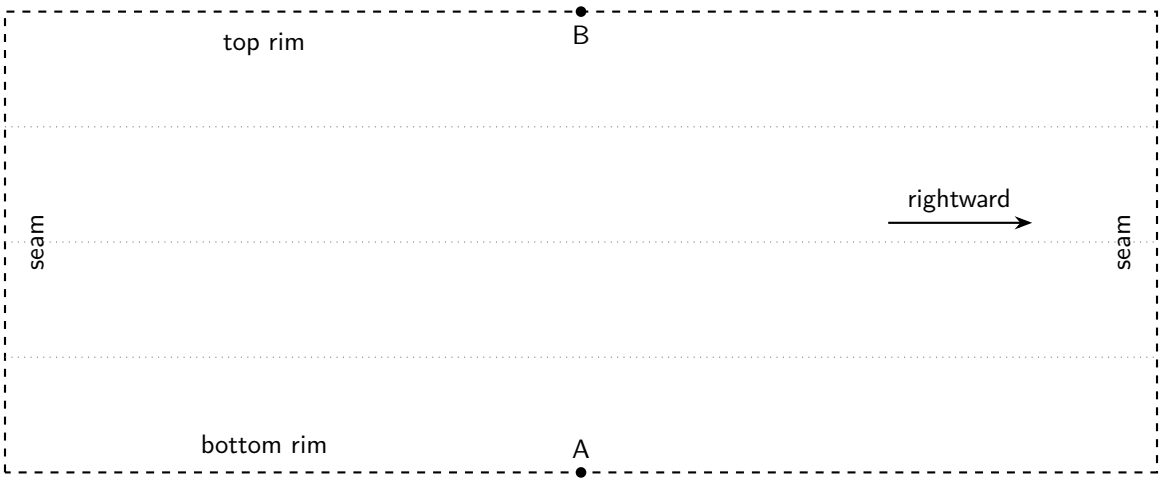
Travel from A to B. At the seam, a rightward crossing counts $+1$; a leftward crossing counts -1. Add these to get the route's *winding*.



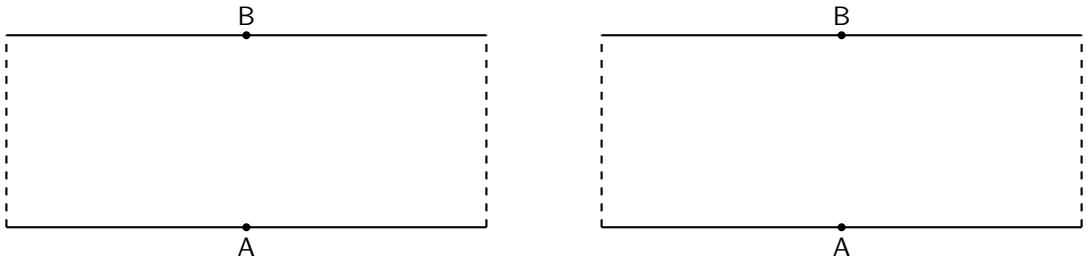
Example: P to Q crosses rightward once. Count $+1$.

Exit and entry stay at the same height.

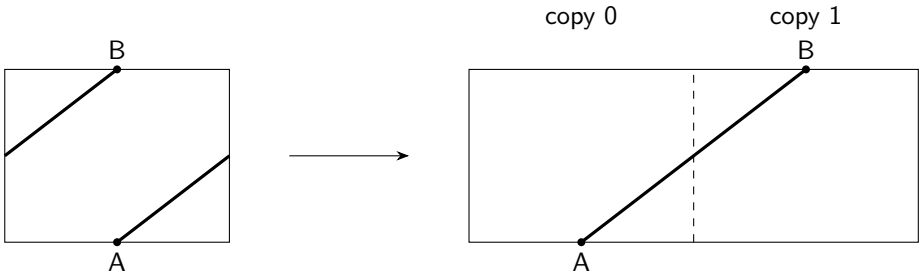
Problem 1: Cut out the rectangle and tape its left and right edges together without overlap. Make yarn routes with winding 0, $+1$, -1 and $+2$. Your partner checks the seam crossings. Make two different-looking routes with the same winding.



Same winding, different routes (winding _____):

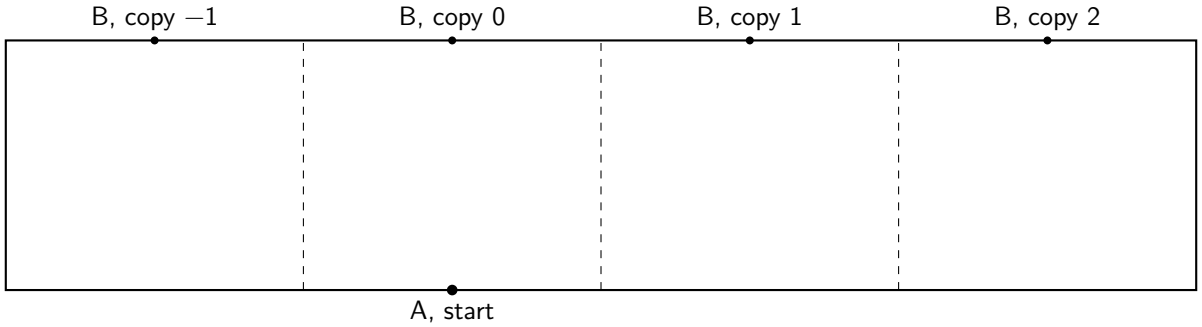


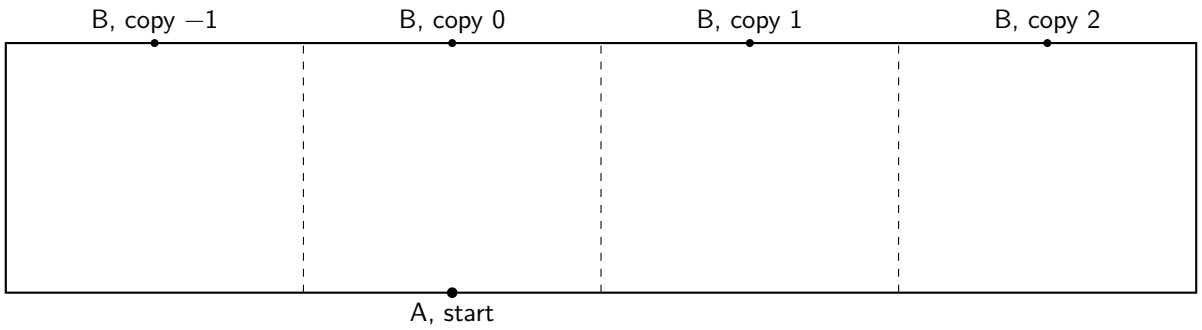
Instead of jumping back across the cut, continue into the next copy of the rectangle. The top dots are copies of the same B on the cylinder.



Example: winding $+1$ ends at B in copy 1.

Problem 2: Draw two different upward routes of winding $+2$, and two of winding -1. Start each at A. Your partner lays each route on the cylinder with yarn.

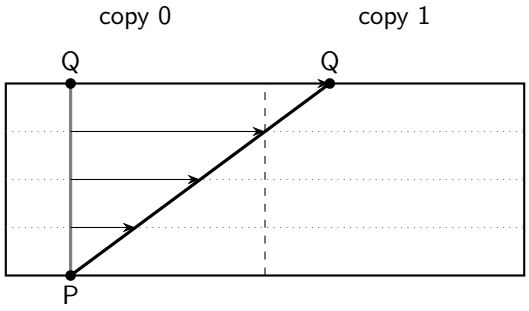




Problem 3: Choose two of your yarn routes. Keeping the endpoints at A and B and the yarn on the cylinder, can you slide one route into the other without crossing itself? Find a pair that can and a pair that cannot. Explain the difference.

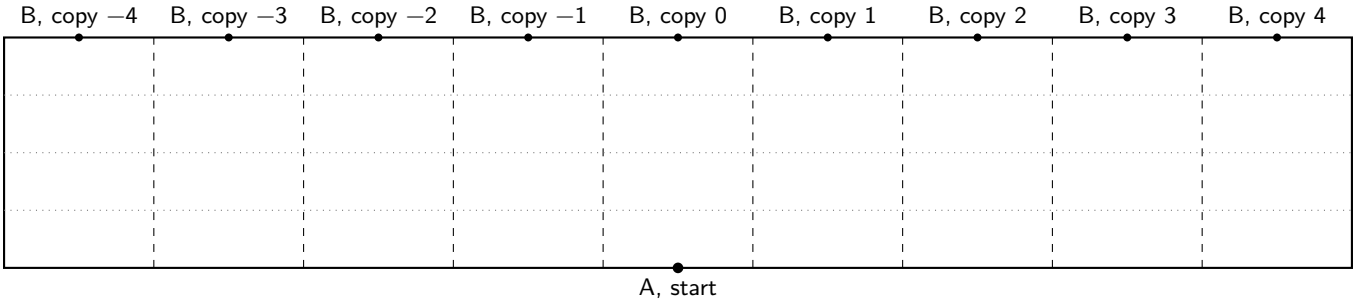
A + twist moves points rightward: none at the bottom, a quarter-turn at the first dotted line, a half-turn at the middle, three-quarter-turns at the next line, and one full turn at the top. Between lines, the shift grows evenly. A − twist uses the same shifts leftward.

Every rim point ends where it started. Draw the changed route and re-lay the yarn to match. The paper cylinder stays still.



Example: one + twist on a straight route from P to Q.

Problem 4: Start with winding 0. Choose a secret word of two to four + and − twists. Make its final route. Your partner tries to make a route with the same winding using the fewest twists. Reveal your word and swap jobs. Which words have the same effect?



Secret word	Ending winding	Shortest word

Problem 5: Find a word of six twists that returns every starting route to its original winding. How can your partner check it without trying every possible starting route?

Problem 6: Lay two yarn routes from A to B with each pair of windings below. Make the number of places where the two strands cross each other as small as possible. Your partner tries to remove another of these crossings without changing either winding.

Both ends are shared. Do not count A or B. Away from the ends, the strands must cross or stay apart: no shared pieces or other touches.

Windings	Fewest strand crossings	Sketch
0 and 1		
0 and 2		
0 and 3		
-1 and 1		
-1 and 2		
2 and 4		

Problem 7: Predict the fewest strand crossings for windings 2 and 7, then for 5 and 5. Find a rule for any pair of windings. Does adding the same twist to both routes change the fewest strand crossings? What can happen if you twist only one?